University of Toronto | Faculty of Information

INF2179H - Personal Challenge

Classification Challenge Report

Instructor	Prof. Mehdi Ataei
Student	Tianyu Wang - 1011076256
Date	March 20, 2026



UNIVERSITY OF
TORONTO

Classification Challenge Report

1. Introduction

The goal of this project is to develop machine learning models to analyze customer reviews from a women's clothing e-commerce dataset. The dataset includes review text, review titles, star ratings, and recommendation indicators. Two classification tasks were addressed in this challenge. The first task is to predict whether a user recommends a product, which is a binary classification problem. The second task is to predict the exact star rating from 1 to 5, which is a multi-class classification problem. According to the assignment, models must be trained using the training dataset and evaluated on the testing dataset.

2. Methodology

For both tasks, the main input was the combined review title and review text. The text was preprocessed by converting it to lowercase, removing non-letter characters, and normalizing extra spaces. The cleaned title and review text were then combined into a single text feature for modeling.

TF-IDF vectorization was used to convert the text into numerical features. This method was chosen because it is effective for representing review text and capturing important words and short phrases related to customer sentiment. Logistic Regression was selected as the classifier for both tasks because it performs well on sparse text data and provides a simple, strong, and reproducible baseline.

For Task 1, the model was trained to predict whether a review recommends a product, and the rating variable was not used, following the assignment requirement. For Task 2, the

model was trained to predict the exact star rating from 1 to 5. A fixed random state was used to support reproducibility.

3. Discussion Question 1: Task 1 result analysis

For Task 1, the best model was used to predict whether a customer recommends a product. According to the assignment, the main evaluation metric for this task is the weighted F1-score, but the report should also clearly present precision, recall, F1-score, and accuracy.

On the testing dataset, the model achieved:

- Accuracy: **0.9038**
- Weighted F1-score: **0.9077**

The classification report shows the following results:

- **Class 0 (not recommended):** precision = **0.69**, recall = **0.86**, F1-score = **0.76**
- **Class 1 (recommended):** precision = **0.97**, recall = **0.91**, F1-score = **0.94**

```
Task 1 - Testing Classification Report
              precision    recall  f1-score   support

           0       0.69      0.86      0.76       851
           1       0.97      0.91      0.94      3847

    accuracy                           0.90      4698
   macro avg       0.83      0.88      0.85      4698
weighted avg       0.92      0.90      0.91      4698

Task 1 - Testing Accuracy: 0.9037888463175819
Task 1 - Testing Weighted F1: 0.9076592177005742
```

These results show that the model performs strongly across both classes, but it performs better on the recommended class than on the not recommended class. For the recommended class, both precision and recall are very high. This means that when the model predicts a review as recommended, it is usually correct, and it is also able to identify most truly recommended reviews.

For the not recommended class, the recall of 0.86 is still relatively strong, which means the model can detect many negative reviews. However, the precision is lower at 0.69, showing that some reviews predicted as not recommended are actually recommended. In other words, the model is better at finding negative cases than at predicting them with the same level of certainty as positive cases.

Overall, the recommendation model performs very well, with especially strong results on recommended reviews. A likely reason is that recommended reviews are more common in the dataset and often contain clearer positive language, which makes them easier for the model to learn. Since the weighted F1-score is above 0.90, the model reaches the full-score threshold for Task 1.

4. Discussion Question 2: Task 2 result analysis

For Task 2, the best model was used to predict the exact star rating from 1 to 5. The official evaluation metric for this task is accuracy, but the classification report also provides useful class-level precision, recall, and F1-scores for understanding which ratings are easier or harder to predict.

On the testing dataset, the model achieved:

- Accuracy: **0.6605**

The classification report shows the following results:

Task 2 – Testing Classification Report

	precision	recall	f1-score	support
1	0.44	0.14	0.22	153
2	0.47	0.23	0.31	317
3	0.48	0.41	0.44	620
4	0.46	0.35	0.40	1017
5	0.75	0.93	0.83	2591
accuracy			0.66	4698
macro avg	0.52	0.41	0.44	4698
weighted avg	0.62	0.66	0.63	4698

These results show that the model performs unevenly across the five rating classes. The easiest rating to predict is 5 stars, which has the highest precision, recall, and F1-score. This suggests that highly positive reviews are the most recognizable, likely because they contain clearer and more consistent positive language patterns.

The hardest ratings to predict are 1 star and 2 stars. Their recall values are especially low, which means the model misses many low-rating reviews. These classes are likely more difficult because they appear less frequently in the dataset and may use more varied or less predictable wording.

The model shows moderate performance on 3-star and 4-star reviews. These ratings are also challenging because they often contain mixed opinions, so the review text is less clearly

positive or negative. Compared with binary recommendation prediction, predicting exact star ratings is more difficult because the model must distinguish among five classes instead of two.

Overall, the star-rating model performs best on clearly positive reviews and struggles more with low and middle ratings. Even so, the overall accuracy of 0.6605 is above the 0.60 full-score threshold, which means the model performs strongly according to the assignment rubric.

5. Real-World Implications

These models could be useful in an e-commerce setting. The recommendation model could help identify likely positive and negative reviews, allowing businesses to monitor customer satisfaction more efficiently. The rating model could help estimate likely star ratings when structured ratings are unavailable.

However, both models also have limitations. They rely only on review title and review text, so they do not use other useful features such as product category, user behavior, or purchase history. In addition, TF-IDF and Logistic Regression work well as baseline methods, but they may not fully capture subtle tone, mixed sentiment, sarcasm, or deeper context.

A practical recommendation would be to use the recommendation model as a support tool for customer feedback monitoring rather than as a fully automated decision system. For future improvement, more advanced language models could be explored to improve performance on difficult rating classes, especially low and middle star ratings.